

AI Prompting Lab Report by Salman Ahmed

Introduction

This lab helped me understand how to communicate with AI more effectively by writing better prompts. I practiced different prompting methods, experimented with changing instructions, and observed how the quality of the responses improved.

1. What I Learned from the Lab

During this lab, I learned that the quality of an AI response depends heavily on the quality of the prompt. Giving clear context, goals, and requirements produces much better answers than asking short or vague questions.

Some of the key lessons I learned include:

- Providing detailed context improves the accuracy of responses.
- AI should not be trusted blindly for recent or changing information without verification.
- Asking AI to think carefully can produce more structured and useful answers for complex decisions.
- Iterating on prompts and giving feedback leads to better final results.
- Different AI models may produce different outputs for the same prompt.
- Image and data analysis features make AI useful beyond simple text generation.

2. How My Prompting Improved During the Process

At the beginning, I asked very simple questions without giving enough details. The answers were often general and not fully useful.

As I continued the lab, I started including:

- Background information
- Constraints and requirements
- Expected output format
- Target audience

This resulted in responses that were more relevant, organized, and closer to what I actually needed.

3. Prompting Techniques That Helped Me the Most

The techniques that I found most useful were:

- **Providing clear context:** Giving complete information helped AI understand my needs better.
- **Using structured prompts:** Breaking requests into goals and expected outputs produced cleaner results.
- **Brainstorm and iterate approach:** Asking for multiple options first and then refining the best one gave higher-quality responses.
- **Requesting balanced comparisons:** Instead of asking AI to agree with an opinion, asking it to compare both sides produced more objective answers.
- **Using "think hard" for difficult decisions:** This encouraged more thoughtful and organized responses.

4. Challenges Faced and How I Solved Them

One challenge was receiving answers that were too generic because my prompts lacked context. I solved this by adding more details and clearly explaining the situation.

Another challenge was understanding when AI might guess instead of providing verified information. I learned to ask for current sources or tell the AI to admit uncertainty if it was not sure.

I also noticed that the first response was not always the best. By refining my prompts and giving follow-up instructions, I was able to get much better results.

Conclusion

Overall, this lab improved my understanding of AI prompting and showed me the importance of writing clear and detailed instructions. I learned that good prompting is an iterative process and that providing the right context leads to more accurate and useful responses. The techniques practiced in this lab will help me use AI more effectively for both daily tasks and professional work.